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New palaeoproterozoic palaeomagnetic data from Central and Northern Finland indicate a long-lived stable position for Fennoscandia

T. Luoto [1,2] J. Salminen,[1,2] S. Mertanen,[2] S.-Å. Elming[3] and L.J. Pesonen[4]

[1]*Department of Geosciences and Geography, P.O. Box 64, 00014 University of Helsinki, Finland. E-mail: toni.luoto@helsinki.fi; toni.luoto@gtk.fi*
[2]*Geological Survey of Finland, P.O. Box 96, 02151 Espoo, Finland*
[3]*Department of Civil, Environmental and Natural Resources engineering, Luleå University of Technology, SE-07187 Luleå, Sweden*
[4]*Department of Physics, University of Helsinki, P.O. Box 64, 00014 , Finland*

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SUMMARY

The Svecofennian gabbro intrusions coincide temporally with the global 2100–1800 Ma orogens related to the amalgamation of the Mesoproterozoic supercontinent Nuna. We provide a new reliable 1891–1875 Ma palaeomagnetic pole for Fennoscandia based on rock magnetic and palaeomagnetic studies on the Svecofennian intrusions in central Finland to fill gaps in the Palaeoproterozoic palaeomagnetic record. By using the new pole together with other global high-quality data, we propose a new palaeogeographic reconstruction at 1885 Ma. This, together with previous data, supports a long-lived relatively stable position of Fennoscandia at low to moderate latitudes at 1890–1790 Ma. Similar stable pole positions have also been obtained for Kalahari at 1880–1830 Ma, Siberia at 1880–1850 Ma, and possibly India at 1980–1775 Ma. A new reconstruction at the beginning of this period indicates the convergence of several cratons at 1885 Ma in the initial stages of the amalgamation of the Nuna supercontinent at low to moderate latitudes. The close proximity of cratons at low to moderate latitudes is further supported by global and regional palaeoclimatic indicators. Stable position of several cratons could indicate a global period of minimal apparent drift at *ca*. 1880–1830 Ma. Before this period, the global palaeomagnetic record indicates large back-and-forth swings, most prominently seen in the high-resolution 2020–1870 Ma Coronation loops of the Slave craton. These large back-and-forth movements have been explained as resulting from an unstable geomagnetic field or basin- or local-scale vertical-axis rotations. However, the most likely explanation is inertial interchange true polar wander (IITPW) events, which is in line with the suggestion of large amplitude true polar wander events during the formation of the supercontinent.

Key words: Europe; Palaeomagnetism; Rock and mineral magnetism; Tectonics and landscape evolution.

1 INTRODUCTION

High-quality palaeomagnetic poles (Buchan *et al.* 1994, 2000) are an essential tool used to reconstruct the palaeogeographic history and relative lateral movement of different continental blocks (i.e. plate tectonics) in geological deep time by establishing the ancient latitudes and azimuthal orientations recorded in rock formations (see review in Pesonen *et al.* 2021). Most of the global high-quality poles are derived from mafic intrusions, which can be precisely dated and are reliable magnetic recorders. However, due to the inherently episodic nature of mafic intrusions, there are large gaps in the record (Evans *et al.* 2021). The majority of the Palaeoproterozoic palaeomagnetic data from Fennoscandia are derived from gabbro intrusions located in Finland and Sweden. Several of these gabbro intrusions have been radiometrically dated and they form age clusters at *ca*. 1885 Ma, 1860 Ma, 1800 Ma and 1780 Ma (poles reviewed in Salminen *et al.* 2021a), but of these, only the 1786 Ma Hoting gabbro provides a key pole (Buchan *et al.* 2000) with a proven primary magnetization (Elming *et al.* 2009). The other Palaeoproterozoic key pole for Fennoscandia was obtained from the 1976 Ma Onega dykes and sills (Lubnina *et al.* 2016). Our aim in this work is to provide a new reliable pole to fill gap at *ca*. 1930–1870 Ma in the palaeomagnetic record of Fennoscandia, targeting *ca*. 1885 Ma gabbros located in central Finland and *ca*. 1866 Ma gabbros located in northern Finland. Some of the *ca*. 1885 Ma gabbros have previously been studied by Pesonen & Stigzelius

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(1972), but their sampling scheme at the time produced only one sample from each gabbro intrusion, which does not fulfil modern statistical requirements and moreover may have not averaged out the palaeosecular variation (Meert *et al.* 2020).

The Svecofennian gabbros temporally coincide with the global 2100–1800 Ma orogens related to the formation of the Mesoproterozoic supercontinent Nuna (e.g. Hoffman 1988, 1989; Zhao *et al.* 2002; Meert & Santosh 2022), also known as Columbia (see Wang *et al.* 2021 for a discussion of supercontinents, megacontinents and supercratons; and Meert 2012 and Evans 2013 for the nomenclature of Nuna versus Columbia). During the transition from Palaeoproterozoic supercratons (Salminen *et al.* 2021b) to the Nuna supercycle (Elming *et al.* 2021), the palaeomagnetic poles of several cratons show large back-and-forth oscillating swings (Mitchell *et al.* 2010; Antonio *et al.* 2017; Gong & Evans 2022). This is most prominently seen in the 2020–1870 Ma Coronation loops obtained from the palaeomagnetic poles of the Slave craton (McGlynn & Irving 1978; Mitchell *et al.* 2010; Gong & Evans 2022). The Coronation loops have been interpreted as products of basin- or local-scale vertical axis rotations (Bingham & Evans 1976; Irving *et al.* 2004; Gong *et al.* 2018), as the result of switching of the geomagnetic field between a dominant axial dipole field and an equatorial dipole field (i.e. as Ediacaran; Abrajevitch & Van der Voo 2010), or by inertial interchange true polar wander (IITPW) events (Mitchell *et al.* 2010; Gong & Evans 2022). IITPW can be considered as a long-term process by which the solid body of the planet shifts relative to its spin axis (Gold 1955). IITPW events are expected in any planet with active geodynamics (Perron *et al.* 2007; Mitchell 2014) and they should have a global effect (Evans 1998, 2003). Mitchell *et al.* (2010) and Antonio *et al.* (2017) also proposed temporally overlapping large swings of poles for Fennoscandia, Amazonia, Superior, Kalahari and India at 1880–1840 Ma, thus supporting the global nature of IITPW. However, the 1880–1840 Ma swings of poles for Fennoscandia proposed by Antonio *et al.* (2017) were later contradicted by Elming *et al.* (2021). In this work, we explore the global *ca*. 2100–1790 Ma high-quality data covering both the proposed IITPW event and its aftermath to examine whether the period of minimal apparent drift, following the period of large back-and-forth movements, was a global phenomenon.

2 GEOLOGY AND SAMPLING

2.1 Kiuruvesi gabbrodiorite (1886 Ma)

The Kiuruvesi area in central Finland is located on the >100-km-wide Raahe–Ladoga tectonic zone, which separates the Svecofennian domain in the southwest from the Archaean Karelian Province in the northeast (Fig. 1; Marttila 1976; Gaál & Gorbatschev 1987). The gneissic basement complex in the area is overlain by the metamorphosed, mostly basic and intermediary volcanites and clastic sediments interbedded by pyroclastic tephra (Marttila 1976). The emplacement of synorogenic hypersthene-bearing dioritic rocks and late orogenic granite intrusions mark the Svecofennian orogeny in the area (Marttila 1981). In addition, the basement complex is cut by aplite, diabase, granite and pegmatite dykes (Marttila 1976). The Kiuruvesi gabbrodiorite is well dated with a U–Pb zircon age of 1886 ± 5 Ma (Kouvo 1978, as cited in Marttila 1981).

A total of 38 samples were taken from five outcrops of the Kiuruvesi gabbrodiorite (Fig. 2) using a portable gasoline field drill. The samples were oriented using both sun and magnetic compasses.

2.2 Ylivieska (1883 Ma) and Mertuanoja gabbros

The Ylivieska area forms part of the accretionary arc complex of central and western Finland (Korsman *et al.* 1997). Grano- and quartzdiorite massifs cover hundreds of square kilometres in the area, but various schists and gneisses are also common (Salli 1961). Some more mafic plutonic rocks with gabbrodioritic or peridotitic composition are also present (Fig. 2; Salli 1961). According to Salli (1961), gabbrodioritic rocks follow the schists conformably, whereas granitoids cut them, indicating that the gabbros are younger than the schists but older than the granitoids (Salli 1961; Pesonen & Stigzelius 1972).

Two separate gabbro units were sampled in the Ylivieska area (Fig. 2). A total of 33 samples were taken with a portable gasoline field drill from six outcrops of the well-dated (U–Pb zircon age 1883 ± 8 Ma; Patchett & Kouvo 1986) Ylivieska gabbro, which covers an area of 45 km^2 (Salli 1961; Pesonen & Stigzelius 1972). A total of 37 samples were taken from eight outcrops of a smaller gabbro unit located near a narrow stream Mertuanoja, northeast of the Ylivieska gabbro (Fig. 2). The samples were oriented using both sun and magnetic compasses.

2.3 Jalokoski gabbro (1866 Ma)

The Jalokoski gabbro is located in Lapland at the border between Finland and Sweden (Fig. 2). This *ca*. 35 km^2 medium-grained gabbroic intrusion is surrounded by monzonites and quartzites (Hiltunen 1982). The Jalokoski gabbro belongs to the well-differentiated igneous Haparanda suite, with rocks ranging from gabbros and diorites to granodiorites (Ödman *et al.* 1949; Ödman 1957; Mellqvist *et al.* 2003). The Haparanda suite is located in the area between the Archaean basement in the northeast and the ∼1880 Ma Svecofennian Skellefte belt in the southwest (Mikkola 1941; Patchett 1978; Väänänen 1984). A nearly concordant U–Pb zircon age of 1866 ± 6 Ma was provided for the Jalokoski gabbro by Väänänen & Lehtonen (2001). The Jalokoski gabbro is clearly foliated, with foliation showing an almost east–west strike and 65° dip to the north (Väänänen 1984).

The Jalokoski gabbro was sampled at five outcrops by Mertanen & Pesonen (1992), with four of the outcrops located in Finland and one in Sweden. A total of 23 samples were collected. Because of the strong magnetic anomaly in the area, only sun-compass directions were used for orientation.

2.4 Nilsiä lamprophyre (1790 Ma) and microtonalite dykes

The Nilsiä study area is located in the northern Savo region in central-eastern Finland. Several microtonalite and lamprophyre dykes intrude the late Archaean (3100–2700 Ma; Sorjonen-Ward & Luukkonen 2005) basement gneiss of the Karelian Province (Fig. 2). These dykes intrude into the large-scale normal fault structures crosscutting early Svecofennian (1890–1875 Ma) regional deformation structures, more or less parallel to the NW–SE-trending palaeosuture zone between the Karelian Province and the Proterozoic Svecofennian domain (Woodard *et al.* 2014). Well-dated lamprophyre dykes (1790 ± 3 Ma; Woodard *et al.* 2014) in the area show no evidence of deformation or metamorphism (Woodard *et al.* 2014). Based on crosscutting relationships, microtonalite dykes predate the lamprophyre dykes in the area, but are younger than the 1860 Ma granite intrusion (Huhma 1981).

Sampling in Nilsiä area (Fig. 2) included a total of 13 samples taken from a lamprophyre dyke on the island of Pieni-Vinkki in

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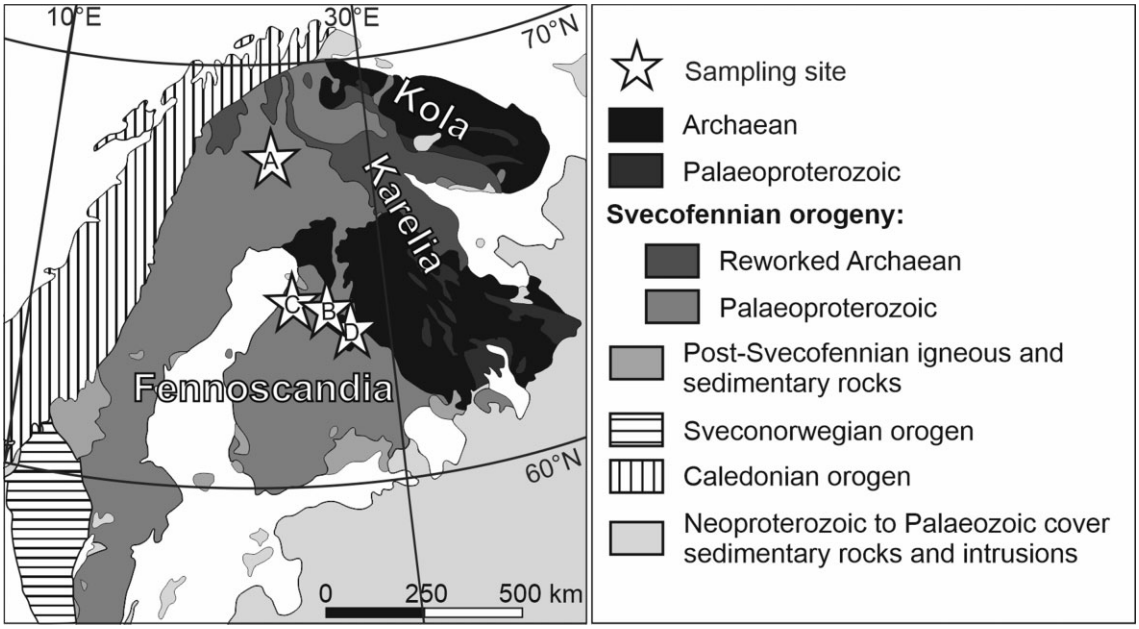


Figure 1. Simplified geological map of the Fennoscandian Shield (modified from Koistinen et al. 2001; Hermansson *et al.* 2008). Sampling sites: A, Jalokoski; B, Kiuruvesi; C, Ylivieska and Mertuanoja; D, Nilsiä.

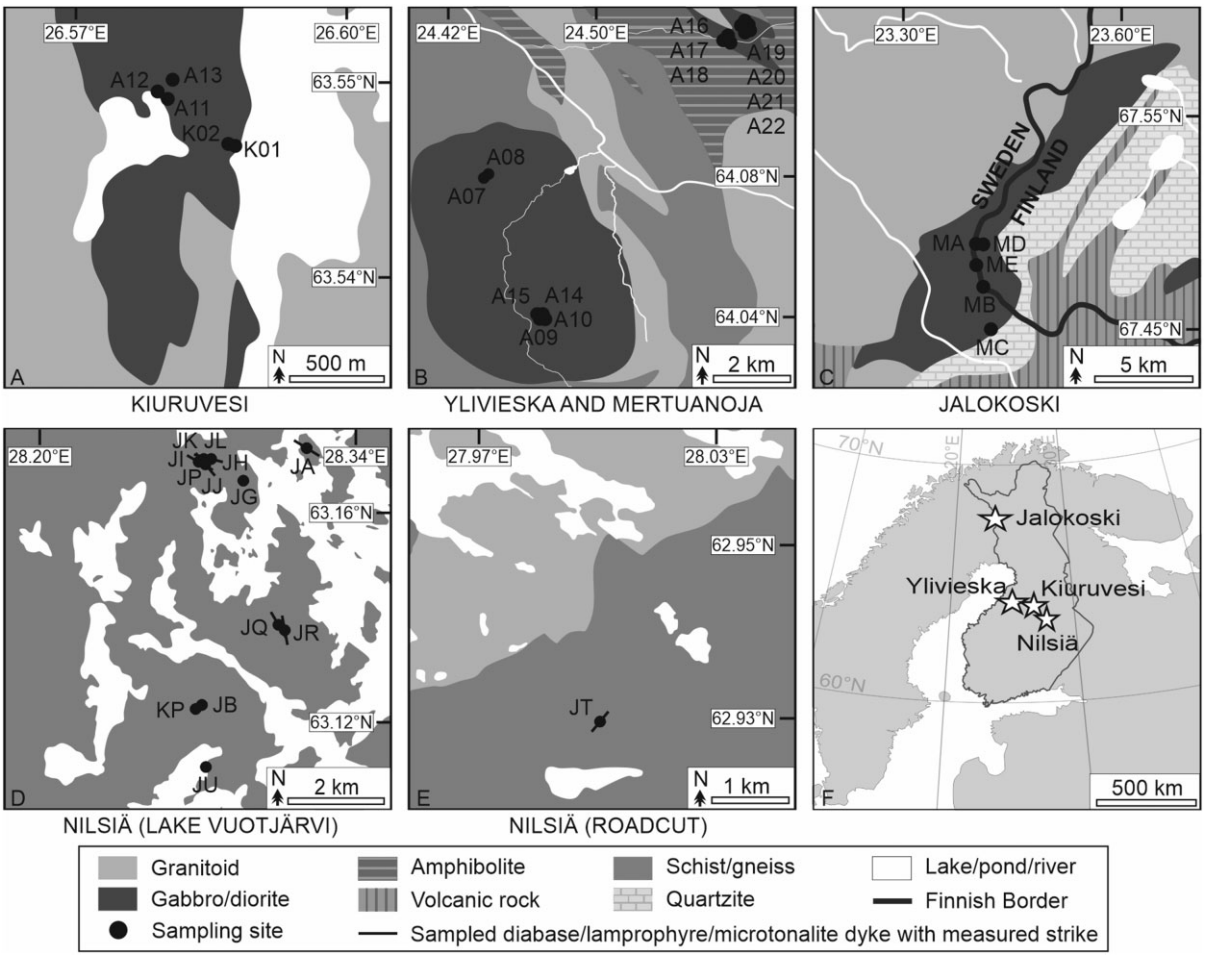


Figure 2. Geological maps of the sampling areas. (a) Kiuruvesi (modified from Marttila 1977); (b) Ylivieska and Mertuanoja (modified from Salli 1955); (c) Jalokoski (modified from Lehtonen 1981; Väänänen 1984; (d) the Lake Vuotjärvi area of Nilsiä lamprophyre and microtonalite dykes (modified from Äikäs 2000) and (e) the roadcut area (Nilsiä microtonalite dyke JT) (modified from Äikäs 2000). (f) An index map showing the sampling areas. Dyke orientations are indicated for Nilsiä dykes.

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lake Vuotjärvi and a total of 74 samples from 13 microtonalite dykes in the surrounding area. Sampling was conducted with a portable gasoline field drill. Samples were oriented using both sun and magnetic compasses.

3 METHODS

The rock magnetic and palaeomagnetic measurements for this study were conducted in the Geophysical Laboratory of the Geological Survey of Finland (GTK) for samples taken from Jalokoski gabbro and Nilsiä, and in the Solid Earth Geophysics Laboratory of the University of Helsinki (UH) for samples taken from the Ylivieska and Kiuruvesi gabbros.

Stepwise alternating field (AF) demagnetizations of the natural remanent magnetization (NRM) of the samples were performed up to a maximum field of 160 mT using a three-axis demagnetizer coupled with a cryogenic 2G DC SQUID magnetometer (UH, GTK), or up to a maximum field of 100 mT using a Schonstedt AF demagnetizer (GTK). Selected sister specimens were thermally demagnetized up to 680 °C using an ASC Scientific model TD-48SC furnace (UH) or a custom-designed furnace (GTK). After each demagnetization step, a 2G DC SQUID magnetometer or an Agico JR4 spinner magnetometer (GTK) was used to measure remanent magnetization. Vector components were visually identified using stereographic and orthogonal projections and demagnetization decay curves (Zijderveld 1967). The mean directions were calculated using the least squares method (Kirschvink 1980). Mean remanent directions were determined for each component using Fisher statistics (Fisher 1953).

The magnetic mineralogy of the samples from the Ylivieska and Kiuruvesi intrusions was investigated using thermomagnetic analyses and hysteresis measurements of selected powdered whole-rock samples. An Agico KLY-3S-CS3 Kappabridge system was used to analyse thermomagnetic properties. During the procedure, the bulk susceptibilities were measured while heating the samples from room temperature to 700 °C and cooling them back to room temperature. Curie temperatures were determined using Cureval 8.0.2 software (http://www.agico.com). Hysteresis properties were measured with a Princeton Measurement Corporation Micromag™ 3900 model vibrating sample magnetometer (VSM) to identify the domain states of the magnetic carriers.

The Lowrie–Fuller test (Lowrie & Fuller 1971) was conducted for selected samples to investigate the domain structure of magnetic minerals. First, the NRM was AF demagnetized in steps up to 160 mT with a cryogenic 2G DC SQUID magnetometer (UH). Next, anhysteretic remanent magnetization (ARM) was imparted with an AGICO LDA3 alternating field demagnetizer, with a peak field of 100 mT and a biasing field of 50 μT in the z-direction. ARM was progressively demagnetized using the same sequence as was used for NRM demagnetization. Finally, an MMPM10 pulse magnetizer was used to induce isothermal remanent magnetization (IRM) in a 1.5 T field along the z-direction. IRM was progressively demagnetized using the same sequence as in the previous steps. The coercivity spectra of NRM, ARM and IRM were then compared. The Lowrie test (Lowrie 1990) was used to examine the different coercivity fractions of IRM of the selected samples from Ylivieska and Kiuruvesi gabbros. IRM was imparted in samples with an MMPM10 pulse magnetizer in three axes: 1.2 T in the z-direction, 0.4 T in the y-direction and 0.12 T in the x-direction. IRM in these three axes is diagnostic for hard, intermediate and soft magnetic carriers, respectively. After inducing IRM, samples were thermally demagnetized in steps up to 680 °C and magnetization along each orthogonal axis was then plotted against temperature.

All palaeomagnetic analyses were performed using Palaeomagnetism.org 2.0 (Koymans *et al.* 2016, 2020). Palaeomagnetic poles were plotted with GPlates 2.2 software (Müller *et al.* 2018). The quality of poles was assessed with the seven-point revised Van der Voo (1990) quality index (R-index) by Meert *et al.* (2020). Averaging of the palaeosecular variation (PSV) of calculated poles was tested with a reliability envelope (Deenen *et al.* 2011, 2014).

4 RESULTS

4.1 Rock magnetic and palaeomagnetic results for the 1886 Ma Kiuruvesi gabbrodiorite

Thermomagnetic curves for selected samples from the Kiuruvesi gabbrodiorite display a significant decrease in magnetic susceptibility at 580 °C, indicating the presence of almost pure magnetite (Fig. 3). We interpret the minor drop in magnetic susceptibility at *ca.* 380 °C to be related to the presence of maghemite, which transforms to magnetite during the heating. Other options could be titanomagnetite, impurities or changes in the magnetic grain size. Heating and cooling curves are irreversible, with cooling curves lacking the lower-temperature phase. The Lowrie–Fuller test indicated that single domain (SD) magnetite is a remanence-carrying mineral (Fig. 4). However, the Lowrie test showed that magnetite is present in soft, intermediate and hard components for the Kiuruvesi gabbrodiorite samples (Fig. 4). In addition, a very narrow hysteresis loop indicates the presence of multidomain (MD) magnetite (Fig. 4).

Palaeomagnetic data from the 1886 ± 5 Ma (Kouvo 1978, as cited in Marttila 1981) Kiuruvesi gabbrodiorite display a characteristic remanent magnetization (ChRM) direction with a NNW declination and an intermediate downward inclination ($D = 343°$, $I = 33°$ with $\alpha_{95} = 3°$ and $k = 403$), reproducing the previous results of Neuvonen *et al.* (1981) from three outcrops (Figs 5 and 6, Table 1). Thermal demagnetization of the specimens produced square-shouldered decay curves with unblocking temperatures in a narrow range below 580 °C, indicating SD magnetite (Fig. 5). AF demagnetization showed typical coercivities of magnetite, with a median destructive field (MDF), which is the demagnetizing field required to halve the magnetization (Dunlop 1983), below 50 mT and no sign of a higher coercivity component with different magnetization directions (Fig. 5). Most of the specimens revealed a viscous component below 10 mT AF or 300 °C, but the direction of this viscous component was highly scattered and it is likely that the viscous component is carried by MD magnetite grains (Fig. 5).

4.2 Rock magnetic and palaeomagnetic results for *ca.* 1885 Ma Ylivieska and Mertuanoja gabbros

Thermomagnetic analyses of the selected samples from the Ylivieska and Mertuanoja gabbros yielded curves similar to the Kiuruvesi gabbrodiorite, although the susceptibility decrease at *ca.* 380° C was significantly steeper in some of the Ylivieska gabbro samples (Fig. 3). The Lowrie–Fuller test demonstrated significantly harder ARM compared to IRM, indicating that magnetization is carried by SD magnetite (Fig. 4). The Lowrie test indicated that

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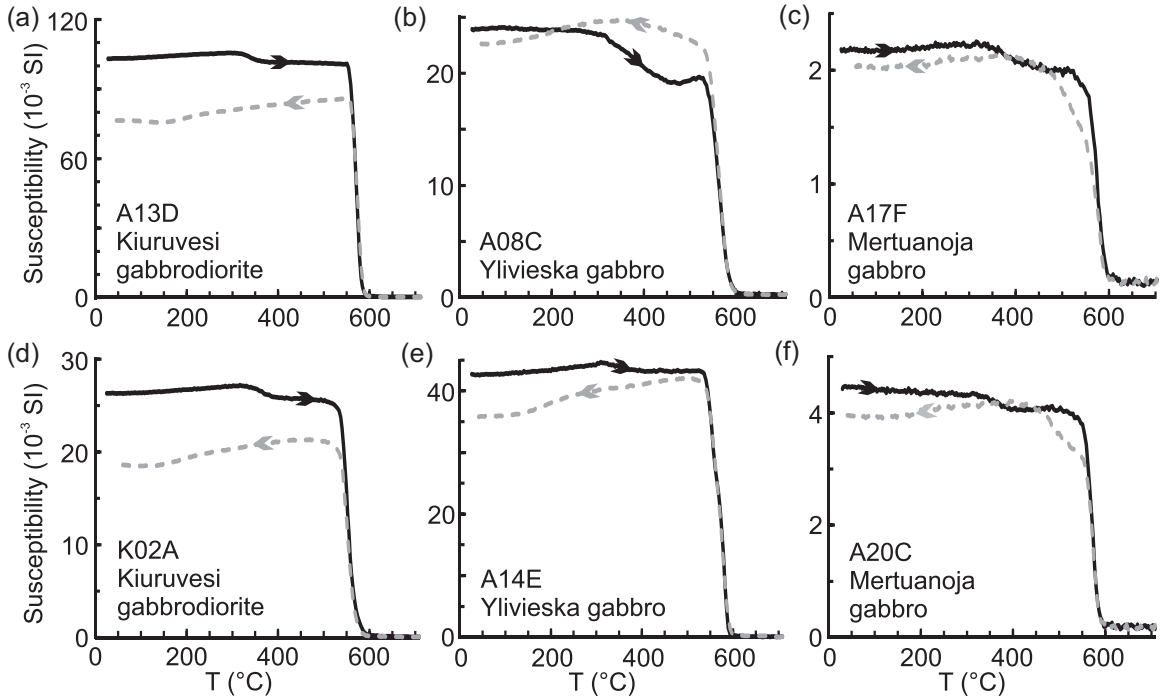


Figure 3. High-temperature thermomagnetic curves for selected samples from Kiuruvesi gabbrodiorite and two gabbro units near Ylivieska. Heating (cooling) curves are drawn with a solid black (dashed grey) line.

magnetite is present in soft, intermediate and hard components for both the Ylivieska and Mertuanoja gabbro samples (Fig. 4). A very narrow hysteresis loop indicates the presence of MD magnetite (Fig. 4).

Palaeomagnetic data from the 1883 ± 8 Ma (Patchett & Kouvo 1986) Ylivieska gabbro display a ChRM direction with a NW declination and intermediate downward inclination (D = 331°, I = 44° with $\alpha 95$ = 7° and k = 49), close to a direction provided by Pesonen & Stigzelius (1972), whose sampling scheme only included one sample from each outcrop (Figs 5 and 6, Table 1). The palaeomagnetic directions from Pesonen & Stigzelius (1972) form two groups, whereas the new palaeomagnetic directions obtained in this study form a single group displaying an elongation in the N–S direction and overlapping these two groups (Fig. 6). Thermally demagnetized specimens showed a square-shouldered shape with unblocking temperatures slightly below 580 °C, indicating SD magnetite (Fig. 5). AF demagnetized specimens displayed typical coercivities of SD magnetite, with an MDF below 50 mT, and no sign of a higher coercivity component with a different magnetization direction. Most of the samples from the Ylivieska gabbro displayed remanence with a viscous component, probably carried by MD magnetite. This component was isolated below 10 mT or 300 °C, with highly scattered directions.

Palaeomagnetic data from the undated Mertuanoja gabbro display a ChRM direction with a NW declination and intermediate downward inclination (D = 339°, I = 44° with $\alpha 95$ = 5° and k = 106), similar to Ylivieska and Kiuruvesi gabbros (Figs 5 and 6, Table 1). The remanent magnetization of the samples from the Mertuanoja gabbro behaved similarly to the samples from the Ylivieska gabbro during thermal and alternating field demagnetization.

4.3 Palaeomagnetic results for the 1866 Ma Jalokoski gabbro

Palaeomagnetic data from the 1866 ± 6 Ma (Väänänen & Lehtonen 2001) Jalokoski gabbro display a ChRM direction with a NW declination and intermediate downward inclination (D = 336°, I = 41° with $\alpha 95$ = 8° and k = 90) (Figs 5 and 6, Table 1). In thermally demagnetized specimens, the decay curves were square-shouldered, with unblocking temperatures in a narrow range below 580 °C, which indicates a low-Ti SD magnetite as the carrier of magnetization (Fig. 5). AF demagnetized specimens displayed typical coercivities of magnetite with an MDF below 50 mT, and no sign of a higher coercivity component with different magnetization directions (Fig. 5). Most specimens also yielded a viscous component revealed after demagnetization below 10 mT AF or 300 °C. This viscous component displayed a direction close to that of the Earth's present field.

4.4 Palaeomagnetic results for the Nilsiä 1860–1790 Ma microtonalite and 1790 Ma lamprophyre dykes

Palaeomagnetic data from 13 microtonalite dykes in the Nilsiä study area display a ChRM direction with a NNW declination and intermediate downward inclination (D = 345°, I = 35° with $\alpha 95$ = 2° and k = 320; Figs 5 and 6, Table 1). Thermally demagnetized microtonalite specimens yielded decay curves with a square-shouldered shape and unblocking temperatures close to 400 °C (Fig. 5). In addition, their behaviour during AF demagnetization was typical for (titano)magnetite (Fig. 5). These curves indicate that the magnetic carrier is probably titanomagnetite with a low to medium Ti content. The dated 1790 ± 3 Ma (Woodard *et al.* 2014) lamprophyre dyke in Nilsiä displayed a ChRM direction with a NNW declination and intermediate downward inclination (D = 343°, I = 41°, $\alpha 95$ =

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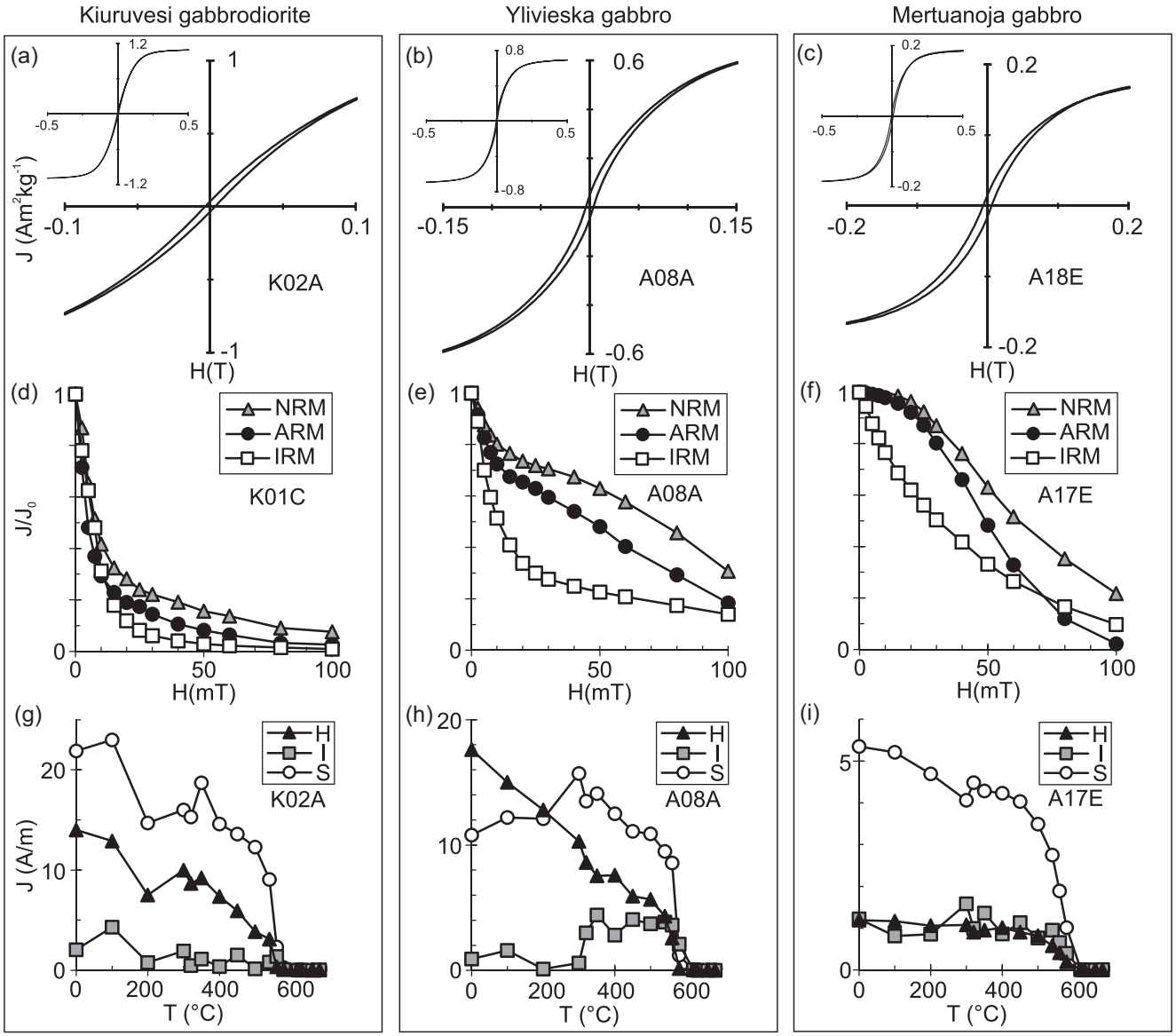


Figure 4. Rock magnetism of selected samples from Kiuruvesi gabbrodiorite and two gabbro units near Ylivieska. Hysteresis curves for (a) Kiuruvesi gabbrodiorite, (b) Ylivieska gabbro and (c) Mertuanoja gabbro; Lowrie–Fuller test (Lowrie & Fuller 1971) for (d) Kiuruvesi gabbrodiorite, (e) Ylivieska gabbro and (f) Mertuanoja gabbro; Lowrie test (Lowrie 1990) for (g) Kiuruvesi gabbrodiorite, (h) Ylivieska gabbro and (i) Mertuanoja gabbro. Abbreviations: NRM, natural remanent magnetization; ARM, anhysteretic remanent magnetization; IRM, isothermal remanent magnetization; H, hard (1.2 T in *z*-direction) component; I, intermediate (0.4 T in *y*-direction) component; S, soft (0.12 T in *x*-direction) component.

4°), which is very similar to the ChRM of the Nilsiä microtonalite dykes.

5 DISCUSSION

5.1 Overview of the palaeomagnetic results

5.1.1 Quality of the data

The obtained magnetization directions of the studied Svecofennian intrusions together with previous data from the same intrusions are presented in Fig. 6. These data, together with site-mean poles and their R-scores (Meert *et al.* 2020), are compiled in Table 1.

The pole calculated for the Kiuruvesi gabbrodiorite fulfils four of the seven revised Van der Voo (1990) criteria (Meert *et al.* 2020). It has a well-determined radiometric age of 1886 ± 5 Ma (Kouvo 1978, as cited in Marttila 1981; R1: 1). The Kiuruvesi mean pole does not statistically average out the PSV, as the observed $A95 = 2°$ is significantly lower than the minimum value $A95_{min} = 5°$ of the reliability envelope (Deenen *et al.* 2011, 2014) (R2: 0). This is also indicated by a precision parameter for pole K = 607 (Fisher 1953). Rock magnetic analysis were used to identify magnetic carriers (R3: 1). Neuvonen *et al.* (1981) reported diabase dykes crosscutting the Kiuruvesi gabbrodiorite, but despite our sampling efforts, none of these intrusions were found in this study. Therefore, it was not possible to perform baked contact tests (Everitt & Clegg 1962; R4: 0). According to Neuvonen *et al.* (1981) and Marttila (1987), the Kiuruvesi gabbrodiorite is crosscut by several subvertical dykes,

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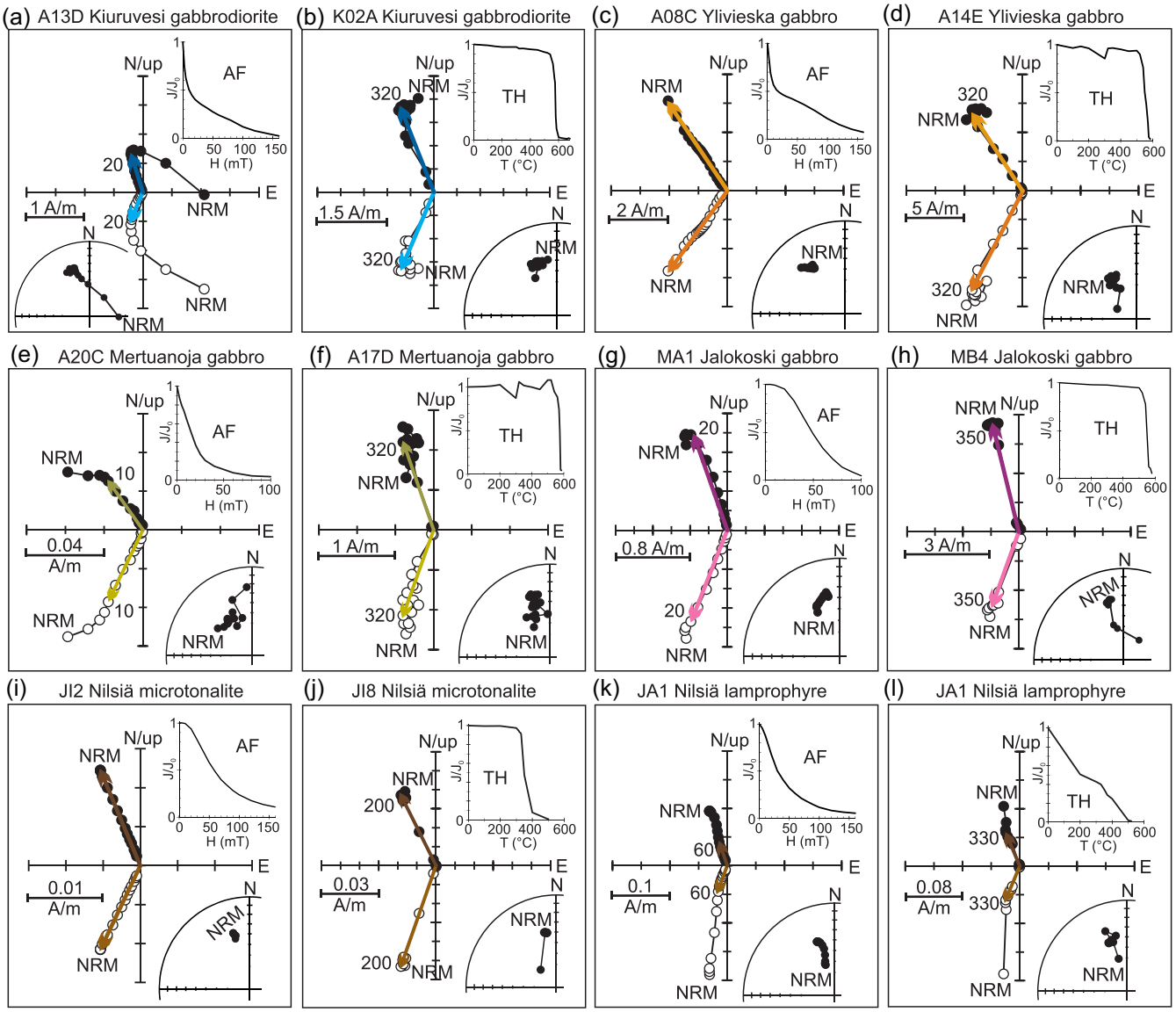


Figure 5. Examples of demagnetization behaviour from alternating field (AF) and thermal demagnetizations (TH). Filled (open) symbols represent the horizontal (vertical) plane in orthogonal projections and downward (upward) directions in stereoplots. Arrows indicate the direction of the characteristic remanent magnetization (ChRM) component. Numbers in orthogonal projections indicate the applied AF field in mT or temperature in °C.

indicating that no tilting has occurred after intrusion (R5: 1). Finally, only normal polarity was observed (R6: 0), and the pole calculated from the Kiuruvesi gabbrodiorite data does not resemble younger poles (R7: 1).

The pole calculated for the Ylivieska gabbro fulfils three of the seven criteria, most importantly having a well-determined radiometric age of 1883 ± 8 Ma (Patchett & Kouvo 1986) (R1: 1). The Ylivieska mean pole averages out the PSV, as the observed $A95 = 6°$ is within the reliability envelope of Deenen *et al.* (2011, 2014; $A95_{min} = 5°$, $A95_{max} = 18°$). However, only six of the eleven studied outcrops have more than three samples, so the Ylivieska data thus do not meet the statistical requirements (R2: 0). Rock magnetic methods were used to identify magnetic carriers (R3: 1), but no palaeomagnetic field tests were performed (R4: 0). Cross-cutting dykes were not observed, so possible tilting cannot be ruled out (R5: 0). Only normal polarity was observed (R6: 0). The pole calculated from Ylivieska gabbro data does not resemble younger poles (R7: 1).

The pole calculated from the Mertuanoja gabbro fulfils only two of the seven R-criteria. It lacks radiometric dating (R1: 0) but is assumed to be related to the intrusion of the Ylivieska gabbro due to its geographical proximity. Eight outcrops were sampled for the Mertuanoja gabbro, with three or more samples from each. In addition, based on the reliability envelope of Deenen *et al.* (2011, 2014), data from the Mertuanoja gabbro average out the PSV ($A95 = 6°$, $A95_{min} = 5°$, $A95_{max} = 22°$), but the precision parameter for the pole (Fisher 1953) is high ($K = 97°$), and the Mertuanoja gabbro data therefore fail criterion R2 (R2: 0). Rock magnetic methods were used to identify magnetic carriers (R3: 1). Due to the lack of crosscutting dykes and contacts to host rock, no palaeomagnetic field tests were performed (R4: 0) and possible tilting could not be ruled out (R5: 0). Re-

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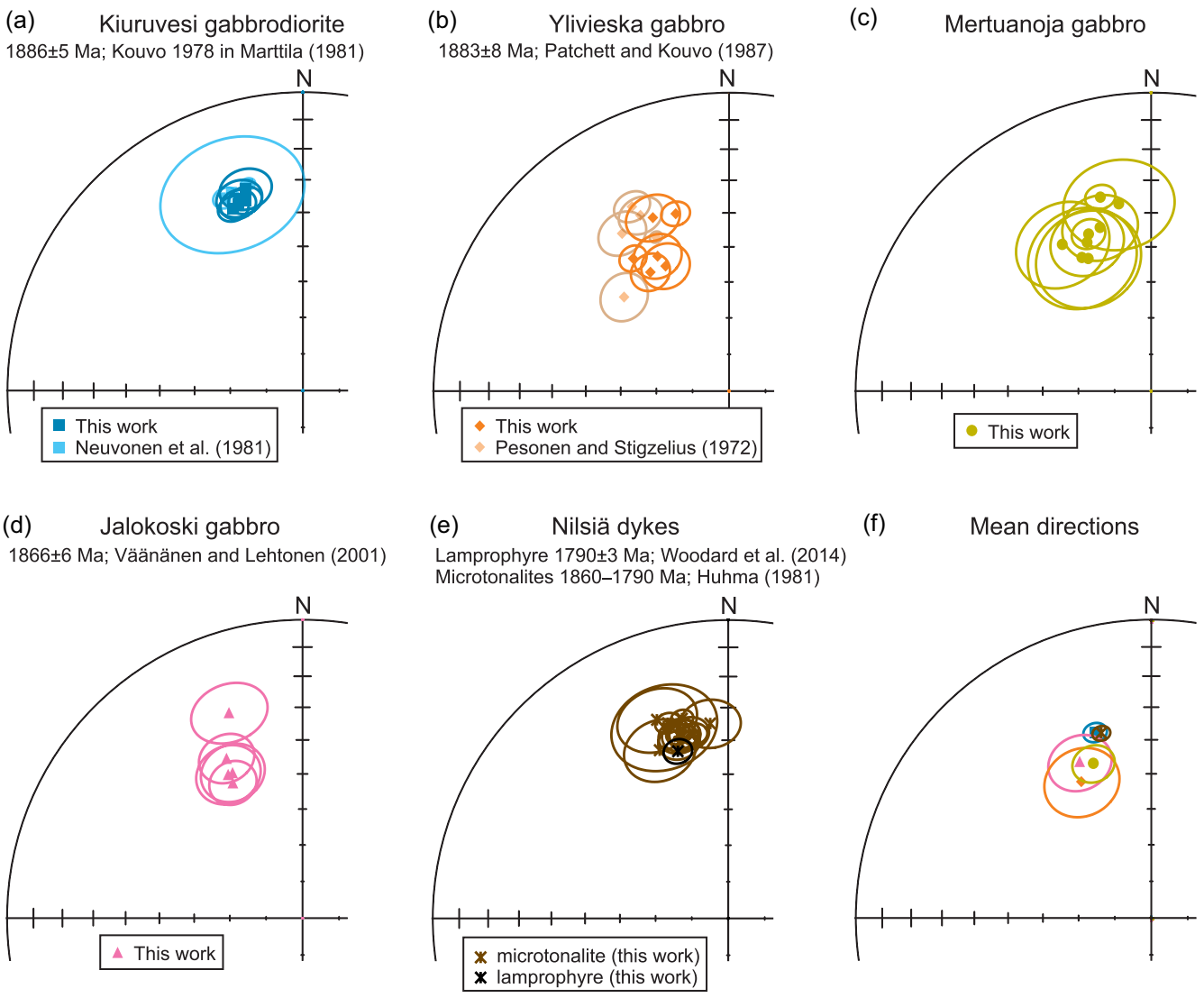


Figure 6. Palaeomagnetic site mean directions for intrusions of Svecofennian age (Table 1): (a) Kiuruvesi gabbrodiorite; (b) Ylivieska gabbro; (c) Mertuanoja gabbro; (d) Jalokoski gabbro and (e) Nilsiä microtonalite and lamprophyre dykes. (f) Mean directions for the different formations. Symbols as in Fig. 5.

versed polarities were not observed (R6: 0). The pole calculated from the Mertuanoja gabbro data does not resemble younger poles (R7: 1).

The Jalokoski gabbro fulfils three of the seven R-criteria, with a well-determined radiometric age of 1866 ± 6 Ma (Väänänen & Lehtonen 2001; R1: 1). Because the data only originate from five outcrops and the precision parameter for the pole (Fisher 1953) is high (K = 139), the pole from the Jalokoski gabbro fails the second R criterion (R2: 0). The AF and thermal demagnetization curves have been used in the identification of magnetic carriers (R3: 1). There were no observations of cross-cutting dykes or contacts to host rock, and therefore no palaeomagnetic field tests were performed (R4: 0) and possible tilting could not be ruled out (R5: 0). Similar to most Svecofennian intrusions, only normal polarity was observed (R6: 0). The pole calculated from the Jalokoski gabbro data does not resemble younger poles (R7: 1).

The pole calculated from the Nilsiä microtonalite dykes fulfils three of the seven R-criteria. Due to the inherited zircon, there are no reliable radiometric ages for these dykes (R1: 0). According to Huhma (1981), the dykes predate the well-dated lamprophyre dykes in the area (1790 ± 3 Ma; Woodard *et al.* 2014), but are younger than the *ca*. 1860 Ma granite intrusion, providing a poorly constrained age estimate of 1860–1790 Ma. The data from the Nilsiä microtonalite dykes do not statistically average out the PSV, as the observed $A95 = 2^\circ$ is significantly lower than the minimum value $A95_{min} = 4^\circ$ of the reliability envelope (Deenen *et al.* 2011, 2014; R2: 0). This is also indicated by an exceptionally high precision parameter for the pole ($K = 328$; Fisher 1953). The shapes of the AF and thermal demagnetization curves have been used in the identification of magnetic carriers (R3: 1). No palaeomagnetic field tests were performed (R4: 0). The 1860–1790 Ma Nilsiä microtonalite dykes are subvertical (Neuvonen *et al.* 1981), as are also the 1790 Ma lamprophyre dykes in the area (Paavola 1984). Moreover, the lamprophyre dykes display no evidence of deformation or metamorphism (Woodard *et al.* 2014). These observations indicate that no tilting has occurred since the dykes intruded (R5: 1). Only normal polarity was observed (R6: 0). The pole calculated from Nilsiä microtonalite data does not resemble younger poles (R7: 1).

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Table 1. Palaeomagnetic results for intrusions of Svecofennian age.

Site	Site name	Lat (N°)	Lon (°E)	(B)/N/n	D (°)	I (°)	α95 (°)	k	Plat (N°)	Plon (E°)	A95 (°)
Svecofennian gabbros											
Kiuruvesi gabbrodiorite (1886 ± 5 Ma; Kouvo 1978 as cited in Marttila, 1981)											
A11	Tuli-Toiviainen 1	63.54900	26.57545	6*/8	345.1	32.1	5.3	160.8	42.8	226.2	4.4
A12	Tuli-Toiviainen 2	63.54918	26.57484	6*/8	343.2	35.4	3.8	314.0	44.7	229.1	3.1
A13	Tuli-Toiviainen 3	63.54982	26.57522	6*/6	341.2	39.6	8.3	66.9	47.6	232.9	7.7
K01	Rytkynjärvi 1	63.54797	26.57595	6*/12	343.9	34.2	7.3	84.8	44.1	227.6	6.3
K02	Rytkynjärvi 2	63.54761	26.57686	10*/19	341.7	34.6	4.3	124.6	43.8	230.8	4.5
7357[A]	Viitaniemi	63.6	26.5	12*/12	345.0	29.0	1.7	652.9	40.8	225.7	?
7359[A]	Rytkyjärvi	63.6	26.5	12*/12	340.0	30.0	17.6	7.0	40.5	232.1	?
7360[A]	Rytky	63.6	26.5	11*/11	338.0	31.0	3.1	217.9	40.7	234.8	?
	MEAN	**63.55**	**26.58**	**8*/69/88**	**342.3**	**33.3**	**2.8**	**403.0**	**43.0**	**229.9**	**2.3**
								(K: 606.5, R: 1010101 4, $A95_{min}$: 5.2°, $A95_{max}$: 22.1°)			
Ylivieska gabbro (1883 ± 8 Ma; Patchett & Kouvo 1986)											
A07	Moortinkallio 1	64.07606	24.44266	5*/7	322.9	52.1	6.3	150.4	51.4	258.8	6.9
A08	Moortinkallio 2	64.07616	24.44271	5*/7	332.3	46.9	6.2	155.6	50.1	243.9	5.7
A09	Honkikallio 1	64.04401	24.47274	5*/7	343.5	36.3	2.1	1321.5	44.8	226.5	2.1
A10	Honkikallio 2	64.04395	24.47291	6*/8	338.1	35.6	7.7	75.7	43.4	233.1	8.2
A14	Honkikallio 3	64.04456	24.47530	6*/8	324.1	49.6	4.2	250.6	49.7	255.7	4.9
A15	Honkikallio 4	64.04484	24.47465	6*/8	332.1	63.4	10.9	38.8	67.2	259.4	13.9
2431/01[B]	Ylivieska	64.03	24.26	1/7*	332.1	30.7	4.5	180.9	38.8	239.4	3.7
2431/02[B]	Ylivieska	64.04	24.27	1/7*	334.5	42.3	1.6	1424.5	47.1	239.4	1.5
2431/03[B]	Ylivieska	64.02	24.29	1/9*	311.7	50.8	6.5	63.7	45.9	270.5	9.2
2431/04[B]	Ylivieska	64.01	24.28	1/5*	325.7	36.2	6.2	153.3	40.5	248.3	5.5
2431/05[B]	Ylivieska	64.05	24.29	1/5*	333.1	34.3	5.5	194.5	41.3	239.0	4.8
	MEAN	**64.04**	**24.38**	**11*/38/78**	**330.6**	**43.7**	**6.6**	**48.6**	**47.7**	**245.7**	**6.4**
								(K: 51.2, R: 1010001 3, $A95_{min}$: 4.6°, $A95_{max}$: 18.1°)			
Mertuanoja gabbro											
A16	Pohjanneva 1	64.10107	24.56574	6*/8	341.4	43.0	9.2	54.5	49.3	231.2	9.9
A17	Pohjanneva 2	64.10081	24.56589	6*/8	346.5	34.9	5.1	175.3	44.4	222.4	4.5
A18	Pohjanneva 3	64.10123	24.56704	6*/8	338.4	43.4	4.2	250.4	48.8	234.8	4.5
A19I	Mertuanjärvi 1	64.11153	24.57890	3*/4	326.5	45.1	16.5	57.0	46.9	251.6	17.5
A19II	Mertuanjärvi 2	64.11153	24.57890	5*/7	334.2	50.3	13.3	34.1	54.6	244.2	15.3
A20	Mertuanjärvi 3	64.11163	24.57883	3*/5	331.9	48.1	15.0	68.5	50.9	245.9	17.5
A21	Mertuanjärvi 4	64.11164	24.57882	4*/6	337.5	45.7	9.0	105.9	50.5	236.6	10.2

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Table 1. Continued

Site	Site name	Lat (N°)	Lon (°E)	(B)/N/n	D (°)	I (°)	α95 (°)	k	Plat (N°)	Plon (E°)	A95 (°)
A22		64.11170	24.57861	4*/6	351.2	36.6	14.3	42.0	46.0	216.8	15.2
	Mertuanjärvi 5										
	MEAN	**64.11**	**24.57**	**8*/37/52**	**338.9**	**43.6**	**5.4**	**106.0**	**49.1**	**234.7**	**5.7**
								(K: 96.6, R: 0010001 2, $A95_{min}$: 5.2°, $A95_{max}$: 22.1°)			
Jalokoski gabbro (1866 ± 6 Ma; Väänänen & Lehtonen 2001)											
MA	Jalokoski 1	67.49	23.40	4*/16	333.5	45.1	7.6	145.7	46.2	238.6	8.4
MB	Jalokoski 2	67.47	23.41	6*/37	334.3	40.2	6.9	96.2	42.7	236.4	7.7
MC	Jalokoski 3	67.45	23.42	3*/13	332.4	47.6	5.9	433.9	47.9	240.7	7.5
MD	Jalokoski 4	67.49	23.41	5*/12	332.2	45.3	8.4	83.9	46.2	240.0	10.0
ME	Jalokoski 5	67.48	23.40	4*/14	343.0	28.2	5.3	300.0	36.4	224.0	3.8
	MEAN	**67.48**	**23.41**	**5*/22/92**	**335.5**	**41.4**	**8.1**	**89.7**	**44.0**	**235.5**	**6.5**
								(K: 138.7, R: 1010001 3, $A95_{min}$: 6.3°, $A95_{max}$: 29.7°)			
Nilsiä lamprophyre dyke (1790 ± 3 Ma; Woodard et al. 2014)											
JA	Pieni-Vinkki	63.18	28.31	13*/45	343.1	41.1	3.6	131.0	48.8	232.1	4.3
Nilsiä microtonalite dykes (1860 – 1790 Ma; Huhma 1981)											
JB	Vuotjärvi	63.12	28.26	3*/4	337.7	38.9	9.1	186.0	46.3	238.9	7.0
JG	Aniaisenmäki 2	63.17	28.27	8*/26	344.7	36.2	3.0	331.0	45.7	229.0	3.3
JH	Aniaisenmäki 3	63.17	28.22	6*/20	345.9	37.1	3.6	355.0	46.5	227.5	3.6
JI	Aniaisenmäki 4	63.17	28.21	7*/17	349.5	37.2	3.2	349.0	47.0	222.8	3.5
JJ	Aniaisenmäki 5	63.17	28.22	5*/15	344.9	36.8	5.2	218.0	46.1	228.8	5.8
JK	Aniaisenmäki 6	63.17	28.22	6*/11	345.4	36.8	4.7	202.5	46.2	228.3	4.8
JL	Aniaisenmäki 7	63.17	28.22	5*/11	344.3	38.5	5.2	221.6	47.2	230.0	5.8
JP	Aniainen	63.17	28.22	5*/11	348.0	36.7	4.9	244.9	46.5	224.6	5.5
JQ	Hipanniemi	63.14	28.30	5*/8	341.3	32.6	13.0	35.4	42.8	233.0	14.3
JR	Hipanniemi 2	63.14	28.30	6*/7	340.1	29.9	9.7	49.0	41.1	233.6	8.8
JT	Tieleikkaus	62.93	28.00	7*/13	342.7	32.0	3.1	378.3	42.9	230.8	3.4
JU	Ruokomäki	63.11	28.26	5*/9	354.7	34.7	7.4	82.0	46.1	215.3	5.1
KP	Kivipuro	63.12	28.26	6*/21	346.9	31.8	2.7	601.7	43.3	225.6	2.3
	MEAN	**63.14**	**28.23**	**13*/74/218**	**345.1**	**35.4**	**2.3**	**319.7**	**45.3**	**228.4**	**2.3**
								(K: 327.8, R: 0010101 3, $A95_{min}$: 4.3°, $A95_{max}$: 16.3°)			

Lat, Lon: site latitude and longitude; B: number of analyzed sites; N: number of analyzed samples; n: number of analyzed specimen; * marks the level used in mean calculations; D: declination; I: inclination; α95: the radius of the 95% confidence cone in Fisher (1953) statistics; k: Fisher (1953) precision parameter; Plat, Plon: latitude and longitude of the virtual geomagnetic pole; A95: radius of the 95% confidence cone of the pole; K: Fisher (1953) precision parameter of pole; R: revised Van der Voo (1990) reliability criteria proposed by Meert et al. (2020) with a R-score (number of criteria fulfilled by the pole) $A95_{min}$, $A95_{max}$: minimum and maximum values of the reliability envelope (Deenen et al. 2011, 2014);
[A]: data obtained from Neuvonen et al. (1981).
[B]: data obtained from Pesonen & Stigzelius (1972).

5.1.2 Origin of the magnetization

The poles from this work, calculated from the Kiuruvesi gabbrodiorite, the Ylivieska gabbro, the Mertuanoja gabbro, the Jalokoski gabbro and the Nilsiä microtonalites, form a spread cluster located at *ca*. 230–250°E and 40–50°N (Fig. 7). The A95 confidence cones of these poles overlap one another with the exception of the Ylivieska pole. However, the Kiuruvesi and Nilsiä poles do not statistically average out the PSV, and therefore may be considered as virtual geomagnetic poles (VGP). According to McElhinny & McFadden (2000), an individual VGP may differ by even 20° from the rotation axis of the palaeomagnetic reference frame, therefore indicating the Kiuruvesi and Nilsiä poles might not be distinguishable from the Ylivieska pole. It is possible that the tight clustering of the studied *ca*. 1885–1790 Ma poles could arise from slow cooling of the large gabbro intrusions combined with minimal drift,

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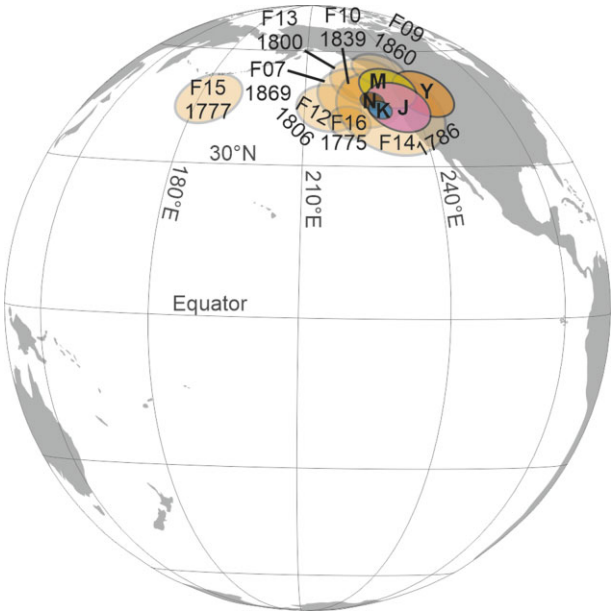


Figure 7. Mean poles from studied formations compared with high-quality 1870–1750 Ma palaeomagnetic poles from Baltica. J: Jalokoski gabbro; K: Kiuruvesi gabbrodiorite; M: Mertuanoja gabbro; N: Nilsiä microtonalites; Y: Ylivieska gabbro.

as magnetization ages *ca*. 10–20 Myr (Buchan *et al.* 2000; Pesonen *et al.* 2003) and even 100 Myr (Elming *et al.* 2009) younger than the crystallization ages of Fennoscandian gabbros have been proposed. However, we consider slow cooling unlikely, as a more recent study of the *ca*. 20-km-long and 4-km-wide Turinge gabbro-diabase showed that it intruded at great depth, indicating relatively rapid cooling with a magnetization age close to the baddeleyite U–Pb age of the rock (Elming *et al.* 2019). In addition, poles calculated in this study overlap with a high-quality palaeomagnetic pole derived from the rapidly cooled *ca*. 1870 Ma Keuruu diabase dykes (Klein *et al.* 2016; Fig. 7), indicating magnetization ages close to this age.

Palaeomagnetic data from Ylivieska gabbro, show a N–S elongation (Fig. 6). According to Beck Jr (1999) elongation could be related to the slow cooling combined with rapid continental drift. However, in the case of the Ylivieska gabbro, this is an unlikely explanation, as tight clustering of 1885–1790 Ma Fennoscandian poles prove against relatively rapid polar wander required for the development of the elongation in response to slow cooling of gabbro. Furthermore, the Lowrie test demonstrates that magnetite is present in three coercivity fractions (Fig. 4). It is probable that the intermediate coercivity fraction has overlapping coercivity spectra with hard coercivity fraction, and therefore secondary component might be incompletely separated, which could explain the observed elongation. However, this direction is not close to commonly observed secondary components (e.g. component B in Mertanen *et al.* 2006b), and therefore this is not a likely explanation, and the cause of elongation remains unsolved. Nevertheless, the pole calculated for the Ylivieska gabbro is within the tight cluster of 1870–1790 Ma Fennoscandian poles (Fig. 7), indicating that the direction of the suggested secondary component does not greatly differ from the direction of the primary magnetization.

We suggest that the magnetization ages are close to the crystallization ages, based on comparisons with the high-quality data and cooling rates of intrusions of approximately the same size. The cause of N–S elongation observed in the palaeomagnetic data from the Ylivieska gabbro remains unsolved, but the mean direction of Ylivieska gabbro seems to be similar to other sites, and therefore considered to present primary magnetization.

5.1.3 A new reliable 1891–1875 Ma palaeomagnetic pole

By combining the data from the similarly aged Kiuruvesi gabbro-diorite and the Ylivieska gabbro (Kouvo 1978, as cited in Marttila 1981; Patchett & Kouvo 1986) and the presumably coeval Mertuanoja gabbro, we calculated a new reliable ∼1885 Ma mean pole at Plat: 47°N and Plong: 238°E (with B: 27, A95: 3°, K: 66; Table 2). Considering that the studied rock units are relatively large gabbro intrusions, each outcrop has been thought to represent a separate cooling unit.

This new mean pole satisfies five of the seven R-criteria (Table 2). It has a well-determined radiometric age of 1891–1875 Ma (R1: 1). The palaeosecular variation is statistically averaged out, as the observed $A95 = 3.5°$ is within the reliability envelope ($A95_{min} = 3.2°$, $A95_{max} = 10.3°$), and there are statistically sufficient number of sites and samples (R2: 1). Rock magnetic methods were used to identify the magnetic carriers (R3: 1). Subvertical dykes that are reported to crosscut the Kiuruvesi gabbrodiabase (Marttila 1987) indicate that no tilting has occurred after the intrusion (R5: 1). The pole does not resemble younger poles (R7: 1).

The new mean pole fills the 60 Myr gap in the palaeomagnetic record of Fennoscandia at *ca*. 1930–1870 Ma. It overlaps with *ca*. 1870, 1860, 1800 and 1790 Ma high-quality poles and *ca*. 1840 Ma and 1810 Ma poor-quality poles of Fennoscandia (Fig. 7), indicating a relatively stable position for Fennoscandia at *ca*. 35–20°N latitudes (Salminen *et al.* 2021a) during the time period from 1885 to 1790 Ma. Coeval occurrences of evaporites, stromatolite-like structures and lateritic palaeosols in Finland and Sweden support these palaeolatitudes (see compilation of palaeoclimatic indicators for Baltica and its subcratons in Salminen *et al.* 2021a).

5.2 The global Palaeoproterozoic palaeomagnetic record

5.2.1 Large global oscillations in palaeomagnetic data at 2060–1880 Ma

During the period of break-up of the proposed Palaeoproterozoic supercratons (Salminen *et al.* 2021b) and the onset of the amalgamation of the Nuna supercontinent (Elming *et al.* 2021), the palaeomagnetic poles of several cratons display large back-and-forth oscillating swings (Fig. 8). This is most prominently seen in the palaeomagnetic poles of the Slave craton of Laurentia at 2020–1870 Ma (McGlynn & Irving 1978; Mitchell *et al.* 2010; Gong & Evans 2022). These swings of up to ∼110° in the apparent polar wander path (APWP) of the Slave craton, called the Coronation loops (McGlynn & Irving 1978), are found in the high-resolution palaeomagnetic record derived not only from mafic intrusions but also from sedimentary rocks from the Great Slave, Coronation, and Kilohigok Basins (Mitchell *et al.* 2010; Gong & Evans 2022). During the Coronation loops, the Slave craton rifted from the proposed Sclavia supercraton (Bleeker 2003; French & Heaman 2010) and amalgamated with other Laurentian Archaean cratons. Such palaeomagnetic data would indicate rapid latitudinal drift and rotation of the Slave craton with plate velocities higher than for modern-type plate tectonics (Zahirovic *et al.* 2015), which therefore argues

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Table 2. Selected global paleomagnetic poles.

Code	Rock unit	Age (Ma)	Age references	Plat (°N)	Plon (°E)	A95 (°)	$R_{1234567}$ Σ	Grade	Pole References
Amazonia									
A1	Armontabo River granite	2080 ± 4	Théveniaut et al. (2006)	-02.7	346.3	15.5	1011001 4	B	Théveniaut et al. (2006)
A2	MEAN Coastal late monzogranites	2060 ± 4, 2069 ± 4, 2061 ± 15, 2095 ± 6	Delor et al. (2003)	-58.5	030.2	5.8	1110010 4	B	Théveniaut et al. (2006)
A3	Surumu volcanics	1966 ± 5, 1956 ± 5	Schobbenhaus et al. (1994); Santos (2003)	-27.4	054.8	9.8	1110101 5	B	Bispo-Santos et al. (2014a)
A4	Santa Rosa – Sobreiro volcanics	1877.4 ± 4.3, 1895 ± 11, 1880 ± 6	Antonio et al. (2017); Pinho et al. (2006)	24.7	139.7	17.7	1111111 7	A	Antonio et al. (2017)
A5	Velho Guilherme suite	1853.7 ± 6.2	Antonio et al. (2017)	31.1	040.1	9.0	1111111 7	A	Antonio et al. (2017)
A6	Colider volcanics	1789 ± 7	Bispo-Santos et al. (2008)	-63.3	298.8	11.4	1110100 4	-	Bispo-Santos et al. (2008)
A7	Avanavero mafic rocks	1788.5 ± 2.5	Bispo-Santos et al. (2014b)	-48.4	027.9	9.6	1111101 6	A	Bispo-Santos et al. (2014b)
A8	Mucajaí complex	1526 ± 2, 1527 ± 2	Heinonen et al. (2012)	-38.2	000.1	12.6	1110111 6	A	Bispo-Santos et al. (2020)
A9	Nova Guarita intrusives	1418.5 ± 3.5	Bispo-Santos et al. (2012)	-47.9	245.9	7.0	1111110 6	A	Bispo-Santos et al. (2012)
Fennoscandia, Kola and Karelia									
F01	Karelian dykes	2446 ± 5	Vuollo & Huhma (2005)	-19.9	278.7	6.1	1110101 5	B	Mertanen et al. (1999)
F02	Varpaisjärvi dolerites/basement	2295 ± 5	Hölttä et al. (2000)	-17.4	262.3	9.8	1111101 6	B	Mertanen et al. (2006b)
F03	Kuetsyarvi formation	2058 ± 6	Melezhik et al. (2007)	24.7	300.8	16.4	1111101 6	B	Torsvik & Meert (1995)
F04	Onega intrusions	1970 ± 3, 1976 ± 9	Lubnina et al. (2017); Kulikov et al. (1999)	44.4	101.5	6.3	1111001 5	A	Lubnina et al. (2017)
F05	Tsuomasvarri gabbrodiorite	1931 ± 2	Mertanen & Pesonen (1994)	40.2	247.3	6.0	1010101 4	B	Mertanen & Pesonen (1994)
F06	MEAN Kiuruvesi / Ylivieska / Mertuanoja gabbros	1886 ± 5, 1883 ± 8	Marttila (1981); Patchett & Kouvo (1986)	46.9	237.5	3.5	1110101 5	*	this work
F07	Keuruu dykes	1867 ± 8, 1870 ± 9	Klein et al. (2016)	46.6	225.3	8.1	0111101 5	B	Klein et al. (2016); recalc: NPW 2022 working group
F08	Jalokoski gabbro	1866 ± 6	Väänänen & Lehtonen (2001)	44.0	235.5	6.5	1010100 3	*	this work
F09	Eastern Murmansk sills	1860 ± 4, 1863 ± 7	Veselovskiy et al. (2019)	54.7	234.7	5.2	1110101 5	B	Veselovskiy et al. (2019)
F10	Haukivesi intrusions	1837–1840	Huhma (1981)	49.0	225.4	3.1	1000101 3	-	Neuvonen et al. (1981)
F11	Nilsiä microtonalite dykes	1860–1790	Huhma (1981)	45.3	228.4	2.3	0010101 3	*	this work
F12	Notträsk gabbro	1806 ± 6	Sadeghi & Hellström (2017)	43.5	216.2	6.1	1010101 4	-	Elming (1985)
F13	Lake Ladoga intrusions	1802 ± 17; 1800 ± 6	Konopelko & Ivanikov (1996)	50.9	229.1	7.2	1010101 4	B	Mertanen et al. (2006a)
F14	Hoting gabbro	1786 ± 10	Hellström & Larson (2003)	43.0	233.3	10.9	1011101 5	A	Elming et al. (2009)
F15	Småland intrusives	1780 ± 3, 1776 +8/-7	Nilsson & Wikman (1997)	45.7	182.7	6.7	1111101 6	A	Pisarevsky & Bylund (2010)
F16	Shoksha fm	1775	Lubnina et al. (2012)	39.7	221.1	4.0	0111111 6	B	Pisarevsky & Sokolov (2001)
F17	Turinge gabbro-diabase	1703 ± 6, 1687 ± 13, 1700 ± 4	Elming et al. (2019)	51.6	220.2	4.8	1011101 5	A	Elming et al. (2019)
F18	Häme and Satakunta dyke swarms	1642 ± 2, 1647 ± 14, 1639 ± 3	Salminen et al. (2017); Luttinen et al. (2022)	26.3	209.1	10.9	1111101 6	A	Salminen et al. (2021c)
F19	Åland dyke swarm	1575.9 ± 3, 1569 ± 3, 1577 ± 12	Salminen et al. (2016)	24.1	192.9	3.3	1111101 6	A	Salminen et al. (2016); recalc: NPW 2022 working group
Kalahari									
K1	Waterberg UBS-I fm	2054 ± 4	Dorland et al. (2006)	36.5	051.3	10.9	1101100 4	A	de Kock et al. (2006)
K2	Bushveld Complex	2054.4 ± 1.3, 2044.1 ± 2.9	Zeh et al. (2015); Scoates & Friedman (2008)	19.2	030.8	5.8	1111110 6	A	Letts et al. (2009)
K3	MEAN Vredefort impact lithologies	2023 ± 4	Kamo et al. (1996)	22.1	046.0	4.0	1111100 5	A	Hart et al. (1995); Carporzen et al. (2005); Salminen et al. (2009); recalc: NPW 2009 working group;
K4	Waterberg UBS-II fm	2050–1930	de Kock et al. (2006)	-10.5	330.4	9.8	0111110 5	B	de Kock et al. (2006)
K5	Hartley LIP	1923 ± 6, 1920 ± 4, 1928 ± 4	Cornell et al. (1998); Alebouyeh Semami et al. (2016)	22.7	328.6	11.7	1110100 4	B	Alebouyeh Semami et al. (2016)

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Table 2. Continued

Code	Rock unit	Age (Ma)	Age references	Plat (°N)	Plon (°E)	A95 (°)	$R_{1234567}$ Σ	Grade	Pole References
K6	Mashonaland dolerites	1871.9 ± 2.2, 1881.1+5.9/-5.1, 1882.7+1.6/-1.5	Hanson et al. (2011); Söderlund et al. (2010)	08.0	338.0	5.0	1111110 6	A	Bates & Jones (1996)
K7	Sand River dykes	1876 ± 68	Morgan (1985)	02.5	009.2	10.1	0110110 4	B	Morgan (1985)
K8	Post-Waterberg older sills	1878.8 ± 0.5, 1871.9 ± 1.2	Hanson et al. (2004)	09.6	013.9	18.0	1110100 4	-	Hanson et al. (2004)
K9	Post-Waterberg dolerite	1872 ± 1	Hanson et al. (2004)	10.3	022.7	13.9	1011100 4	-	de Kock et al. (2006)
K10	Black Hill Dike Swarm (BHDS)	1871-1839	Olsson et al. (2010)	15.3	014.9	5.6	0111110 5	*	Djeutchou et al. (2021)
K10a	BHDS episode 1	1873 ± 1	Djeutchou et al. (2021)	11.8	010.6	11.8	1110111 6	*	Djeutchou et al. (2021)
K10b	BHDS episode 2	1860 ± 2	Djeutchou et al. (2021)	26.8	013.3	19.3	1010101 4	*	Djeutchou et al. (2021)
K10c	BHDS episode 3	1848 ± 2	Djeutchou et al. (2021)	12.5	013.9	22.8	1011101 5	*	Djeutchou et al. (2021)
K11	Soutpansberg Basin volcanics	<1831	Geng et al. (2014)	23.3	016.6	21.0	0010100 2	-	Hanson et al. (2004); Gose et al. (2006)
K12	Post-Soutpansberg sill province	<1831	Geng et al. (2014)	26.3	017.0	9.2	0010100 2	-	Hanson et al. (2004)
Superior (in eastern reference frame)									
Su1	Fort Frances	2076+5/-4	Buchan et al. (1996)	42.8	184.6	6.1	1110100 4	A	Halls & Heaman (2000); recalc: Evans & Halls (2010)
Su2	Lac Esprit	2069 ± 1	Buchan et al. (2007)	62.0	170.5	6.4	1110101 5	A	Buchan et al. (2007); recalc: Evans & Halls (2010)
Su3	Minto	1998+4/-1.3, 1998 ± 2	Buchan et al. (1998); Hamilton et al. (2016)	38.7	171.5	13.1	1010110 4	A	Buchan et al. (1998); recalc: Evans & Halls (2010)
Su4	Molson B+C2	1877+7/-4, 1884 ± 2	Halls & Heaman (2000)	28.9	218.0	3.8	1111111 7	A	Halls & Heaman (2000); recalc: Evans & Halls (2010)
Su5	Haig/Flaherty/ Sutton Mean	1870.3 ± 0.7, 1870.1 ± 1.1	Hamilton et al. (2009)	01.0	245.8	3.9	1111110 6	B	Schmidt (1980); Schwarz et al. (1982); recalc: NPW 2009 Working Group
Slave/Laurentia									
S1	Indin dykes	2126 ± 3	Buchan et al. (2016)	36.0	284.0	7.0	1111111 7	A	Buchan et al. (2016)
S2	Lac de Gras dykes	2023 ± 2, 2027 ± 4	Buchan et al. (2009)	11.8	267.9	7.1	1111101 6	A	Buchan et al. (2009)
S3	Hackett fm.	1969 ± 1	Bowring & Grotzinger (1992)	-04.7	285.1	12.4	1010100 3	*	Gong & Evans (2022)
S4A	Rifle (Western River) fm	1963 ± 6	Bowring & Grotzinger (1992)	14.0	341.0	7.7	1111110 6	B	Evans & Hoye (1981)
S4B	Rifle fm.	1963 ± 6	Bowring & Grotzinger (1992)	-04.2	287.5	5.9	1110100 4	*	Gong & Evans (2022)
S5	Link fm.	1921 ± 11	Bowring & Grotzinger (1989)	-14.5	275.0	9.5	1110100 4	*	Gong & Evans (2022)
S6	Ghost Dike swarm	1887 ± 5	Buchan et al. (2016)	02.0	254.0	6.0	1111110 6	A	Buchan et al. (2016)
S7	MEAN Seton / Akaitcho / Mara	1885 ± 5	Davis & Bleeker (2007)	-06.0	260.0	4.0	1111111 7	B	Mitchell et al. (2010)
S8	MEAN Kahochella, Peacock Hills	1882 ± 4	Bowring & Grotzinger (1992); Mitchell et al. (2010)	-12.0	285.0	7.0	1100111 5	B	Mitchell et al. (2010)
S9	Tochatwi fm.	1885–1870	Mitchell et al. (2010)	-18.0	216.0	12.0	1001110 4	-	Evans & Bingham (1976)
S10	Stark fm.	1885–1870	Mitchell et al. (2010)	-15.0	212.0	8.0	1110110 5	-	Bingham & Evans (1976)
S11	Brown Sound fm.	1885–1870	Mitchell et al. (2010)	-14.0	242.6	13.6	1010100 3	*	Gong & Evans (2022)
S12	Kuuvik fm.	1885–1870	Mitchell et al. (2010)	-22.6	279.8	5.5	1010100 3	*	Gong & Evans (2022)
S13	Douglas Peninsula fm., Pethei group	1870 ± 4, 1882 ± 4	Davis & Bleeker (2007)	-18.0	258.0	14.2	1010111 5	B	Irving & McGlynn (1979)
S14	Takiyuak Formation	1870 ± 4, 1882 ± 4	Davis & Bleeker (2007)	-13.0	249.0	8.0	1110111 6	B	Irving & McGlynn (1979)
S15	Mean Pearson A / Peninsular sill / Kilohigok basin sill	1870 ± 4	Mitchell et al. (2010)	-22.0	269.0	6.0	1111111 7	A	Mitchell et al. (2010)
L1	Dubawnt group	1785 ± 4	Rainbird & Davis (2007)	07.0	277.0	8.0	1111110 6	B	Park et al. (1973)
L2	Cleaver dykes	1740 +5/-4	Irving et al. (2004)	19.4	276.7	6.1	1011101 5	A	Irving et al. (2004)
L3	Melville Bugt diabase dykes[1]	1622 ± 3, 1629 ± 1, 1632 ± 1, 1635 ± 3	Halls et al. (2011)	02.7	261.4	8.7	1110101 5	B	Halls et al. (2011)
L4	Western Channel diabase	1592 ± 3, 1590 ± 4	Hamilton & Buchan (2010)	09.0	245.0	6.6	1011101 5	A	Irving et al. (1972)

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Table 2. Continued

Code	Rock unit	Age (Ma)	Age references	Plat (°N)	Plon (°E)	A95 (°)	$R_{1234567\ \Sigma}$	Grade	Pole References
Siberia									
Si1	Lower Akitkan Khibelen group, Malaya Kosa fm	1849 ± 11, 1875 ± 14, 1864 ± 11, 1878 ± 4, 1877.0 ± 3.8, 1870.7 ± 4.0	Poller et al. (2004); Didenko et al. (2009a); Didenko et al. (2009b)	-30.8	098.7	3.5	1011101 5	A	Didenko et al. (2009a); Didenko et al. (2009b)
Si2	Upper Akitkan group	1863.0 ± 8.7, 1863 ± 9	Didenko et al. (2009a); Didenko et al. (2009b)	-22.1	097.5	5.2	1011101 5	A	Didenko et al. (2009a); Didenko et al. (2009b)
Si3	Shumikhinsk granitoids	1844 ± 1, 1844 ± 15, 1855 ± 5	Shcherbakova et al. (2006); Larin et al. (2003)	-24.9	106.6	5.2	1010011 4	B	Didenko et al. (2003)
Si4	Elgetey fm	1732 ± 4	Didenko et al. (2013)	-07.1	003.5	13.2	1011111 6	A	Didenko et al. (2015)
Si5	Ulkan granite	1729.7 ± 1.9, 1724.7 ± 3.6	Didenko et al. (2010)	-42.1	069.4	4.4	1110001 4	B	Didenko et al. (2015)
Si6	Kuonamka dykes	1503 ± 5	Ernst et al. (2016)	-06.0	054.0	19.8	1010111 5	B	Ernst et al. (2000)
Si7	West Anabar intrusions	1502 ± 2, 1503 ± 2	Evans et al. (2016)	-25.3	061.4	4.6	1110101 5	A	Evans et al. (2016)
India									
I1	Dharwar-Bastar-SGT	2367 ± 1	Meert et al. (2021)	13.0	062.0	4.6	1111111 7	A	Meert et al. (2021)
I2	Dharwar-Bastar	2256 ± 5, 2257 ± 4, 2252.7 ± 0.5, 2252 ± 9	Nagaraju et al. (2018a)	13.0	116.0	14.0	1110101 5	B	Nagaraju et al. (2018a);recalc: Meert et al. (2021)
I3	Dharwar-Bastar	2216.6 ± 0.9	Nagaraju et al. (2018b)	34.0	124.0	6.6	1110110 5	B	Meert et al. (2021)
I4	Dharwar-Bastar	2205.2 ± 0.8, 2208.1 ± 0.8, 2206 ± 4, 2212.3 ± 0.8, 2212.9 ± 0.4, 2207 ± 0.7	Nagaraju et al. (2018a)	51.0	108.0	9.2	1111101 6	A	Meert et al. (2021)
I5	Dharwar-Bastar	2081.8 ± 1.1	Kumar et al. (2015)	41.0	184.0	4.6	1111101 6	A	Meert et al. (2021)
I6	Bundelkhand	1979.1 ± 7.9	Pradhan et al. (2012)	58.0	309.0	4.8	1110111 6	B	Meert et al. (2021)
I7	Dharwar-Bastar	1886.2 ± 5.8	Parashuramulu et al. (2021)	34.0	334.0	4.5	1111101 6	A	Meert et al. (2021)
I8	South India	1766 ± 1, 1764.5 ± 0.9	Shankar et al. (2018)	43.0	320.0	10.3	1111101 6	A	Meert et al. (2021)

Code refers to codes used in Figs 7, 8 and 9; Plat, Plon: pole latitude and longitude; A95: 95% confidence circle of the pole; $\mathbf{R_{1234567\ \Sigma}}$: revised Van der Voo (1990) reliability criteria proposed by Meert et al. (2020) with a R-score (number of criteria fulfilled by the pole); Grade: grade of the pole. A, B: high-quality poles, ∗: not graded yet, - lower-quality pole.
[1]Greenland rotated to Laurentia according to Euler pole (ELat: 67.5°, ELong: 241.5°, ERot: −13.8°) of Roest & Srivastava (1989).

against apparent polar wander alone being able to explain the observed swings.

Competing models have been proposed to explain the Coronation loops: local vertical axis rotation of small blocks of the Slave craton (Irving *et al.* 2004; Gong *et al.* 2018), an unstable magnetic field involving a considerable horizontal dipole (i.e. as Ediacaran; Abrajevitch & Van der Voo 2010) and inertial interchange true polar wander (IITPW; Mitchell *et al.* 2010; Gong & Evans 2022). However, using high-resolution sedimentary data from the Goulburn supergroup of the Slave craton, Gong & Evans (2022) noted that although basin-scale rotations were able to bring some correlative poles closer, some other poles were simultaneously displaced even further apart. Therefore, although basin-scale rotations may affect some poles, they alone cannot explain the observed back-and-forth movement. Moreover, antipodal directions present in the high-resolution Slave craton data indicate that the geomagnetic field should not have been significantly different from a geocentric axial dipole (GAD) field during the Coronation loops (Gong & Evans 2022). Gong & Evans (2022) summarized these results by suggesting that the IITPW probably controls the overall pattern of the Coronation loops.

Large swings in high-quality Palaeoproterozoic palaeomagnetic data have been obtained globally, including for the Amazonia, Kalahari, India and Superior cratons (Mitchell *et al.* 2010; Antonio *et al.* 2017). Antonio *et al.* (2017) also proposed a large 1880–1850 Ma swing for Fennoscandia, but this was later contradicted by Elming *et al.* (2021), who only used poles with proven primary magnetization for Fennoscandia in their analyses, unlike Antonio *et al.* (2017). Here, we extend the time period with reliable poles to cover the duration of the Coronation loops and explore the global high quality 2060–1870 Ma palaeomagnetic data.

Fennoscandian palaeomagnetic data (including the Kola and Karelia subcratons) indicate a *ca*. 180° distance between the 2060 Ma Kuetsyarvi pole from Kola (Torsvik & Meert 1995) and the 1980 Ma Onega pole from Karelia (Lubnina *et al.* 2017), and a *ca*. 150° distance between the Onega pole and the 1930 Ma Tsuomasvarri pole (Mertanen & Pesonen 1994) (Fig. 8). Data obtained from the Kalahari craton show a *ca*. 80° distance between the 2020 Ma mean pole from the Vredefort impact lithologies (Hart *et al.* 1995; Carporzen *et al.* 2005; Salminen *et al.* 2009) and the 1990 Ma pole from the Waterberg unconformity-bounded sequence II (de Kock *et al.* 2006), and a *ca*. 35° distance between the 1880 Ma

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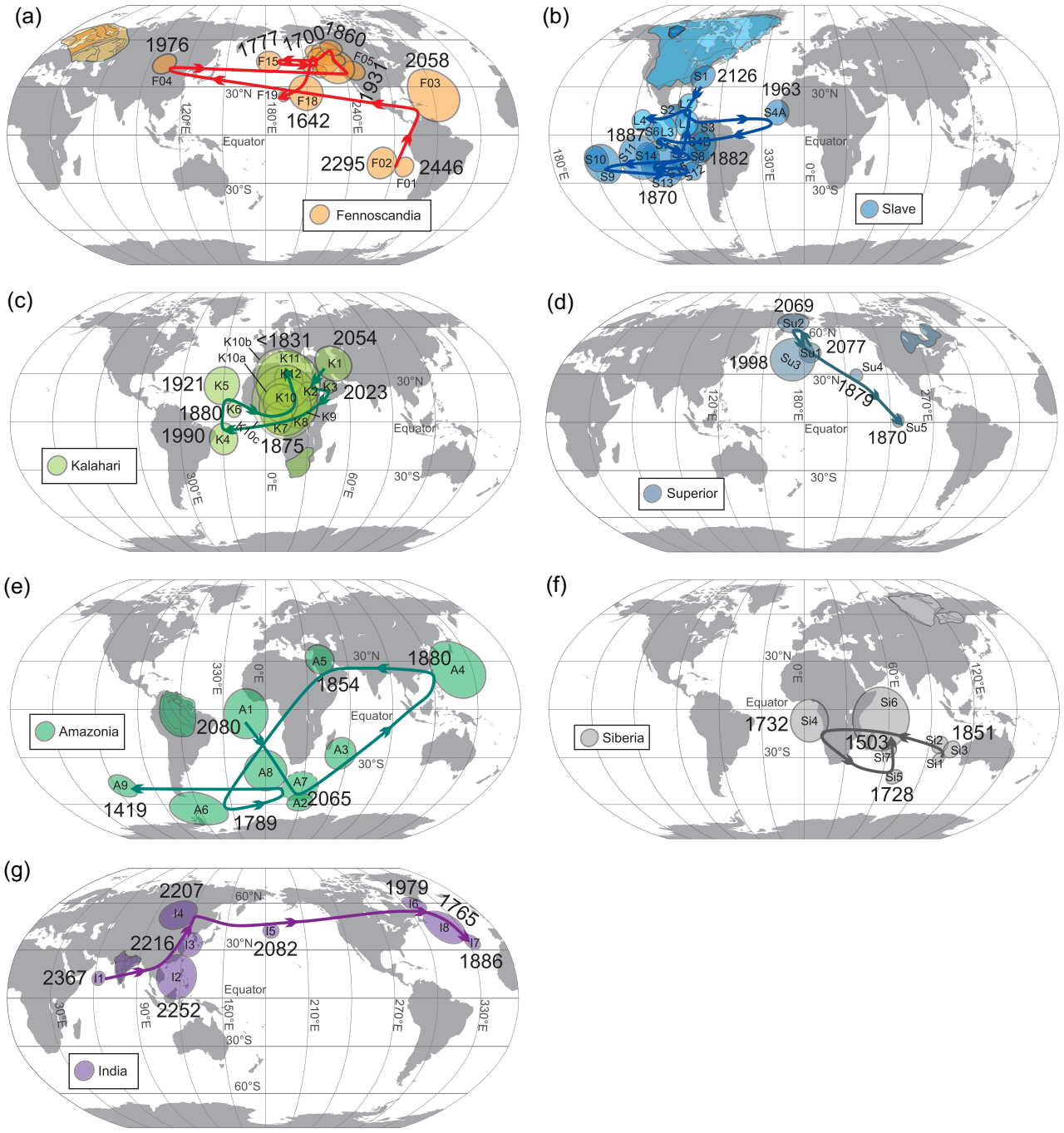


Figure 8. Apparent polar wander paths at *ca*. 2100–1500 Ma. (a) Fennoscandia (including Kola and Karelia); (b) Slave; (c) Kalahari; (d) Superior; (e) Amazonia; (f) Siberia and (g) India. Data and pole codes are given in Table 2.

pole from the Mashonaland dolerites (Bates & Jones 1996) and the 1875 Ma pole from the post-Waterberg older sills (Hanson *et al.* 2004). Palaeomagnetic data from the Superior craton indicate a *ca*. 40° distance between the *ca*. 2000 Ma Minto pole (Buchan *et al.* 1998; Evans & Halls 2010) and the 1880 Ma Molson pole (Halls & Heaman 2000; Evans & Halls 2010). Palaeomagnetic data from the Amazonian craton are scarce, but show relatively large distances between poles at 2080–1880 Ma: *ca*. 60° at 2080–2065 Ma, and *ca*. 120° at 2065–1880 Ma (Fig. 8). Similar to the Amazonian craton, data for the Indian craton are scarce, but large distances between the poles are obtained at 2220–1980 Ma, indicating an apparent drift of *ca*. 180° (Fig. 8).

In summary, the back-and-forth swings in the 2060–1880 Ma APWPs of the Fennoscandia, Slave, Amazonia and Kalahari cratons form hairpin shapes with different lengths and ages of their apexes. In addition, high-quality poles from India indicate significant drifting without back-and-forth movement. We believe that this may be due to a lack of coeval and high-resolution data and propose that the oscillation may indicate IITPW events (Evans 2003; Gong & Evans 2022). This is in line with the suggestion of large-amplitude

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TPW events during the formation of a supercontinent (Evans 1998, 2003; Mitchell *et al.* 2012, 2021; Mitchell 2014).

5.2.2 A period of stable pole positions in the aftermath of a proposed IITPW event

High-quality 1890–1790 Ma palaeomagnetic data for Fennoscandia show the previously mentioned tight cluster of poles between longitudes of 210–240°E, indicating a period of stable pole positions after the significant drifting at 2060–1930 Ma (Fig. 8). Djeutchou *et al.* (2021) compiled high-resolution 1890–1830 Ma palaeomagnetic data for the Kalahari craton from the Black Hills dyke swarm and coeval regional magmatic provinces (Fig. 8, Table 2), indicating minimal apparent drift for the Kalahari craton at 1880–1830 Ma, similarly to Fennoscandia (Figs 7 and 8). In addition, Djeutchou *et al.* (2021) provided 1890–1830 Ma magnetostratigraphy, displaying periods of normal, reversed and mixed polarity. One polarity reversal is shown at *ca*. 1870–1860 Ma, correlating well with the only known reversed-polarity data from Fennoscandia between 1890 and 1830 Ma, recording dual polarity in the *ca*. 1870 Ma Keuruu dykes (Klein *et al.* 2016). In addition, dual polarities are observed in the Superior craton from the *ca*. 1880 Ma pole obtained from the Molson dykes (Halls & Heaman 2000; Evans & Halls 2010) and the *ca*. 1870 Ma Haig–Flaherty–Sutton mean pole (Schmidt 1980; Schwarz *et al.* 1982). The Kalahari data also indicate reversed polarity at *ca*. 1830 and 1850 Ma (Djeutchou *et al.* 2021), but no high-quality palaeomagnetic data are available for Fennoscandia during those times. Correspondingly, the *ca*. 1840 Ma Haukivesi intrusions indicate a normal polarity period for Fennoscandia at that time, but no 1840 Ma data exist for the Kalahari craton.

Siberian data are scarce, but three *ca*. 1880–1850 Ma palaeomagnetic poles are within 10° of each other, and show reversed or dual polarities, therefore supporting stable palaeopole locations (Fig. 8). The palaeomagnetic data from the Indian craton are too scarce to draw conclusions, but three high-quality 1980–1770 Ma poles may indicate a relatively stable position for India during that period (Fig. 8). In contrast, two high-quality palaeomagnetic poles from Amazonia show a *ca*. 100° distance between 1880 and 1850 Ma (Antonio *et al.* 2017).

5.3 A palaeogeographic reconstruction at 1885 Ma

We provide a new palaeogeographical reconstruction at 1885 Ma, at the beginning of the suggested period of minimal apparent drift following the proposed IITPW event (Fig. 9). The reconstruction largely follows the 1860 Ma reconstruction provided by Elming *et al.* (2021), with equatorial positions for India, Siberia, Sarmatia and Volgo-Uralia, and Fennoscandia and the Superior craton at moderate latitudes. At that time, the Slave craton had already amalgamated with Rae and Hearne cratons (Hoffman 2014), and is reconstructed to low to moderate latitudes, further east of Superior, with the Manikewan Ocean separating these cratons (Stauffer 1984). Contradicting the reconstruction by Elming *et al.* (2021), we prefer a position of Kalahari in the northern hemisphere following the 1880 Ma reconstruction by Djeutchou *et al.* (2021). Djeutchou *et al.* (2021),, based this position for Kalahari on their magnetostratigraphic correlations with coeval reversed polarity data from Fennoscandia and Superior, and the alignment of the Black Hills and Mazowe dyke swarms of the Kalahari craton with the *ca*. 1880 Ma Circum-Superior large igneous province magmatic centre of the superior craton (Fig. 9), originally proposed by Minifie et al. (2013).

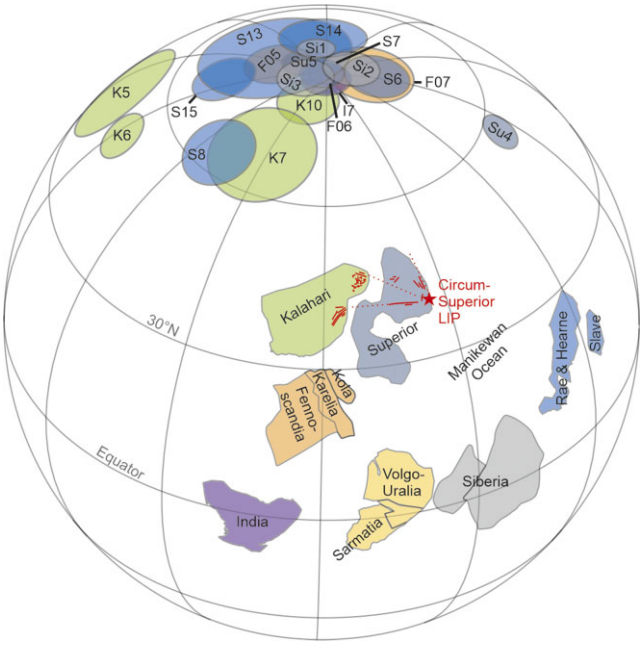


Figure 9. Palaeogeographic reconstruction at 1885 Ma (modified from Djeutchou *et al.* 2021 and Elming *et al.* 2021). Euler parameters [Elat (°), Elong (°), Erot (°)] to absolute: Superior (−39.14°, 021.62°, −118.27°); Slave (−34.12°, 038.01°, −126.47°); Fennoscandia (57.28°, −174.57°, 085.97°); Sarmatia (54.98°, 138.78°, 121.7°; Elming *et al.* 2021); Siberia (30.3°, 130.4°, 209.3°; Elming *et al.* 2021); India (16.05°, 098.61°, 133.70°) and Kalahari (47.07°, −028.06, 108.24°). In addition, within the block rotation for Siberia: Aldan to Anabar (60.00°, 115.00°, 025.00°; Evans 2009). For Kalahari and Superior, 1.89–1.83 Ga magmatism is indicated with red markings.

Global low to moderate latitudes are supported by various types of palaeoclimatic evidence at the global and regional scale. The abundance of global black shale deposits and banded iron formations at *ca*. 2000–1700 Ma indicates the presence of huge landmasses at low to moderate latitudes (Condie *et al.* 2001). This is further supported by the increase in stromatolite deposits observed in the marine carbonates of North America (Peters *et al.* 2017), the extensive *ca*. 2000–1800 Ma red bed successions in the Kalahari region (Dorland *et al.* 2006; Moen 2006; Geng *et al.* 2014; van Niekerk & Beukes 2019; de Kock *et al.* 2021) and the *ca*. 1890–1840 Ma evaporites, stromatolite-like structures and lateritic palaeosols observed in Fennoscandia (Lager 2001; Lahtinen & Nironen 2010; Dahlin *et al.* 2012).

Together with the previous 1870–1790 Ma palaeomagnetic data (e.g. Elming *et al.* 2021; Salminen *et al.* 2021a), this new reconstruction supports a long period of minimal apparent drift for Fennoscandia at low to moderate latitudes. Several cratons had already assembled together at this time, possibly explaining the period of stable positions.

6 CONCLUSIONS

(i) We provide a new reliable 1891–1875 Ma pole for Fennoscandia based on our rock magnetic and palaeomagnetic study on the Svecofennian Kiuruvesi, Ylivieska and Mertuanoja gabbros. In addition, two lower-quality 1866 Ma and 1860–1790 Ma poles were obtained from the Jalokoski gabbro and the Nilsiä microtonalites, respectively. These complement the previous Svecofennian data

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for Fennoscandia reviewed in Salminen *et al.* (2021a) and verify that Fennoscandia was located at low to moderate latitudes at 1885–1790 Ma, further supported by the occurrences of evaporites, stromatolite-like structures and lateritic palaeosols in Finland and Sweden at that time.

(ii) Unlike the previous studies on Kiuruvesi (Neuvonen *et al.* 1981) and Ylivieska gabbros (Pesonen & Stigzelius 1972), the new palaeomagnetic data meets the present-day statistical requirements for high-quality data. New data are also statistically shown to average out the palaeosecular variation.

(iii) The palaeomagnetic data from cratons in Fennoscandia, the Kalahari, Siberia and possibly India indicate a global period of minimal drifting at *ca*. 1880–1830 Ma, following the period of large craton movements at *ca*. 2060–1880 Ma. We agree on the previous interpretations of these movements being related to the IITPW events (Mitchell *et al.* 2010; Antonio *et al.* 2017; Gong & Evans 2022), thought to be related to the amalgamation of supercontinent (Evans 1998, 2003; Mitchell *et al.* 2012, 2021 Mitchell 2014).

(iv) The new 1885 Ma palaeogeographic reconstruction at the beginning of this period indicates that most of these cratons were located at low to moderate latitudes at this time, further supported by global and regional palaeoclimatic indicators. Reconstruction shows that Kalahari, Fennoscandia, India, Siberia, Sarmatia, Superior and Slave cratons were at close proximity. Observed stable positions of Fennoscandia, Kalahari, Siberia and possibly India could therefore indicate that these cratons were drifting as a stable joint landmass, after the amalgamation indicated by the proposed Palaeoprotorozoic IITPW event.

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AUTHOR CONTRIBUTION

Research was conceptualized by Johanna Salminen and Toni Luoto. Sampling was mainly conducted by Toni Luoto, Johanna Salminen and Satu Mertanen. Satu Mertanen also provided results from earlier unpublished studies. Toni Luoto processed the data and analysed the results. The work was supervised by Johanna Salminen. Toni Luoto wrote the original paper. Johanna Salminen, Satu Mertanen, Sten-Åke Elming and Lauri Pesonen reviewed and edited the paper.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY

Site level data are available in the paper. Sample/specimen level data available by request from corresponding author.

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